Parallax Fire

Manual

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NASA Space Apps Challenge 2026: Harmonization of MODIS and VIIRS Hot Spots

Data snapshot parallax-20261010T190934Z-ce88f489, created 2026-10-10 19:22 UTC

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1. What Parallax Fire is

Parallax Fire shows how many fire detections two NASA satellites recorded, month by month, for 3 checked places (Bangladesh, New South Wales, California) and the divisions and states inside them. The data runs from 2001 to 2025.

The two satellites do not count the same way. The older one (MODIS) sees squares about 1 km wide. The sharper one (VIIRS) sees squares about 375 m wide, so it notices smaller fires. When it joined in 2012, counts leapt up even though no more land burned. Parallax Fire lines the two up on one scale so the years can be compared fairly.

What it does not do

- It does not predict future fires. It estimates what the sharper satellite would probably have counted in the past.
- A detection is a hot pixel seen from space. It is not a confirmed wildfire.
- Missing is never shown as zero. Estimates are always labelled and drawn dashed.

Data snapshot: `parallax-20261010T190934Z-ce88f489`, created 2026-10-10 19:22 UTC. Whole-snapshot result: **Raw comparison only**.

2. Quick start

Home	A short scroll story of the idea. Pick a place and go.
Story	One real month, March 2024 in Bangladesh, in a few steps with its real numbers.
Calendar	Pick a place. Each square is one month. Click a square to see its summary.
Map	Where inside the place the satellites saw fire in the month you picked.
Two satellites	The older and the sharper satellite on one chart, with the 2012 jump shown raw and on one scale.
Live	The Earth with today's satellite fire detections and where the satellites are now.
Results	How well the estimate held up on unseen years, place by place.
Data	Where every number comes from, downloads (CSV, printable brief), snapshot facts.

The place, year and month you pick are shared across pages and kept in the address, so a link opens the same view.

3. Reading the calendar

Rows are years, columns are months. Colour shows the count of fire detections: darker means more. Each colour is paired with a symbol so the calendar also reads without colour.

fire detection	A spot where a satellite saw a pixel hotter than its surroundings. It is a hot pixel, not a confirmed wildfire.
Estimate (dashed)	What the sharper satellite would probably have counted. It is worked out from the older satellite, so it is an estimate, never a measurement.
no data for this month	Nothing was recorded for this month. Missing is never shown as zero. Hatched squares mean no data.
part of the month only	The satellite record covers only part of this month, so the count may be too low.
unusually high / low for the season	Compared with the same month in other years, this month sits near the top or the bottom.

Partial months in this snapshot

Country	Aqua partial month	S-NPP partial month
Australia	2002-07	2012-01
Bangladesh	2002-07	2012-01
United States	2002-07	2012-01

Partial months show their observed counts but are left out of learning, the simple rules and the unusual-month flags.

4. Two satellites and the jump

MODIS looks at the ground in squares about 1 km wide. It has been flying since 2000 and 2002 (Terra and Aqua). VIIRS sees squares about 375 m wide, so it notices smaller fires. It has been flying since 2012.

Gluing the two records together makes the line leap upward in 2012. The "Two satellites" page shows that jump twice: raw, and on one scale (the estimate).

The chart has a switch between **squeezed** and **even** scale. Squeezed gives every tripling the same height, so quiet and huge months can be read together, and zero stays at the bottom. Even gives equal steps equal counts, which is honest about size but makes quiet months look tiny.

The lines

- **Older satellite:** what MODIS counted. Terra and Aqua are shown separately; Terra is never mixed into the estimate.
- **Sharper satellite:** what VIIRS counted, since 2012.
- **Estimate (dashed):** what the sharper satellite would probably have counted, with a shaded likely range.
- **Fixed heat sources:** industry, kilns and similar, counted separately.

5. How the estimate is made

1. **Count the hot pixels.** For each place and month we count the fire detections of each satellite.
2. **Notice the jump.** The sharper satellite sees smaller fires, so counts leap when it joins.
3. **Learn the translation.** In 2013–2020 both satellites were flying. We study how many detections the sharper one made whenever the older one saw a given number. These are the *years used to learn* (34,803,042 cell-months).
4. **Try several methods and pick one.** Each was compared on 2021–2022, years held back for choosing (10,972,320 cell-months). The winner is the **Tree model**.
5. **Check it on unseen years.** 2023–2024 were kept sealed until the end and used once (10,972,320 cell-months). The results in chapter 6 come from those years only.
6. **Add a likely range.** Next to every estimate we show a likely range (8 in 10). Chapter 6 measures how often the real value landed inside it.
7. **Fill in the past.** For 2003–2012, when only the older satellite existed, we show the estimated past, always dashed and labelled.

The methods compared

Error is the average miss of the monthly total over the whole pool of checked places, on the years used to choose (lower is better). A method only counts if it beats the simple rules by at least 10% (the gate rule).

Method	Average miss	Passes the bar?
Copy the older satellite	3,686	no
Simple ratio by month	1,218	no
Statistical model	1,190	no
Tree model	589	yes
Neural net	601	yes

The chosen method is the passing one with the lowest miss: **Tree model** (technical id `B2_xgb`).

Years at a glance

estimated past (before 2012)	2003–2012
years used to learn	2013–2020
years used to choose	2021–2022

years used to check	2023–2024

6. Results by place

Every row is measured on the unseen check years (2023–2024). "Average miss" is how far off the monthly total was, on average, in fire detections; lower is better. A place is labelled **Checked on unseen years** only if the estimate beat both simple rules and its likely range was within 10% points of the 80% aim (each place has 24 checked months, so the coverage is uncertain by about 8%).

Places checked on their own

Place	Result	Average miss, ours	Simple rule: copy older satellite	Simple rule: monthly ratio	Real value inside likely range	Months checked
Bangladesh	Checked on unseen years	98 (beats both)	627	109	75%	24
New South Wales	Raw comparison only	375 (beats both)	4,087	1,640	100%	24
California	Checked on unseen years	794 (beats both)	2,219	909	83%	24

The whole snapshot is labelled **Raw comparison only** because it takes the result of its weakest checked place: New South Wales (interval coverage outside 0.80±0.10).

Reported separately (for information; they do not decide the label)

These sit inside a larger checked place, or are whole countries dominated by huge counts.

Place	Result	Average miss, ours	Simple rule: copy older satellite	Simple rule: monthly ratio	Real value inside likely range	Months checked
Barishal	not checked on its own	2	4	3	50%	24
Chattogram	not checked on its own	69	501	94	50%	24
Dhaka	not checked on its own	13	39	25	67%	24
Khulna	not checked on its own	12	20	19	33%	24
Rajshahi	not checked on its own	19	40	27	42%	24

Place	Result	Average miss, ours	Simple rule: copy older satellite	Simple rule: monthly ratio	Real value inside likely range	Months checked
Rangpur	not checked on its own	8	8	7	33%	24
Sylhet	not checked on its own	7	16	13	46%	24
Australia	not checked on its own	8,552	102,204	12,745	100%	24
United States	not checked on its own	4,456	24,857	3,616	92%	24

Source: `metrics.json` in snapshot `parallax-20261010T190934Z-ce88f489`, fields `release.regions` and `release.reported_separately`.

7. Live data and the Earth view

The Live page uses real, current feeds only. Each layer shows its source, a time and an honest state: loading, none in the area, or feed unavailable.

Fire detections, last 24 hours	NASA FIRMS near-real-time files for MODIS, NOAA-20 VIIRS and NOAA-21 VIIRS, worldwide. Fetched through a server function and cached about 15 to 30 minutes.
Satellite imagery	NASA GIBS true-colour images of the day before, and night lights.
Satellites now	Terra, Aqua, Suomi NPP, NOAA-20 and NOAA-21, calculated in your browser from public orbit elements (CelesTrak), refreshed no more often than every 2 hours.
Basemap	OpenFreeMap (OpenStreetMap data). If it cannot load, a plain outline map is used and the page says so.

What "near-real-time" means

These detections arrive within a few hours of the overpass and are not final. They are not part of the monthly calendar, which uses the yearly FIRMS country files.

Suomi NPP feed

At the time of checking, NASA's Suomi NPP near-real-time file contained only its header and no detections. The page then says "none reported" for that satellite. It never shows an empty feed as zero fires.

Ground cameras are not embedded because their licences do not allow it. The satellite imagery is the "camera", and is labelled as imagery.

8. Data, sources and licences

Source	Files	Years	Fetched
Older satellite (MODIS, Aqua and Terra), NASA FIRMS yearly country files Australia, Bangladesh, United States	75	2001–2025	2026-10-10
Sharper satellite (VIIRS on S-NPP), NASA FIRMS yearly country files Australia, Bangladesh, United States	42	2012–2025	2026-10-10
Country and state outlines, Natural Earth	2	n/a	2026-10-10

Licences and credits

- **Fire detections:** NASA FIRMS (firms.modaps.eosdis.nasa.gov). We acknowledge the use of data and imagery from LANCE FIRMS operated by NASA's Earth Science Data and Information System (ESDIS).
- **Satellite imagery:** NASA EOSDIS GIBS / Worldview.
- **Satellite orbits:** CelesTrak (celestrak.org).
- **Basemap:** OpenFreeMap, OpenMapTiles, data from OpenStreetMap contributors.
- **Country and state outlines:** Natural Earth, public domain.
- **Bangladesh districts:** geoBoundaries (gbOpen, CC BY 3.0 IGO), source BBS / OCHA ROAP.

FIRMS acknowledgement: "We acknowledge the use of data and/or imagery from NASA's Fire Information for Resource Management System (FIRMS) (https://earthdata.nasa.gov/firms), part of NASA's Earth Observing System Data and Information System (EOSDIS)."

Downloads

The Data page downloads every month of the place and years you picked as a spreadsheet (CSV) and opens a one-page printable brief. Estimates stay labelled and missing months stay missing.

9. Limits and honest use

- It does not say what will happen next. It lines up counts from the past and the present.
- A detection is a hot pixel seen from space. It is not a confirmed wildfire and gives neither size nor damage.
- Clouds and smoke can hide fires. FIRMS gives no count of cloud-free looks, so we cannot say how many fires were hidden.
- The estimate cannot recover small fires the older satellite never saw. It tells you what the sharper satellite would typically have counted, with a range.
- Results hold per place. A good result for one place says nothing about another.
- With 24 checked months per place, the results are coarse evidence (uncertain by about 8% on coverage).
- Fixed heat sources such as industry are counted separately. Crop and prescribed burning cannot be told apart from other vegetation fires and are included.
- Months with no satellite record in this snapshot (Aqua / Terra / S-NPP): Australia 18 / 0 / 132; Bangladesh 18 / 2 / 132; United States 18 / 0 / 132. The S-NPP figure includes every month before it flew.
- This snapshot is real data, not demo data.

10. Glossary

Plain word	Meaning	Technical term
place	The area you are looking at: a whole country, or one division or state inside it.	AOI (area of interest)
fire detection	A spot where a satellite saw a pixel hotter than its surroundings. It is a hot pixel, not a confirmed wildfire.	active-fire pixel
older satellite (MODIS)	MODIS looks at the ground in squares about 1 km wide. It has been flying since 2000 and 2002 (Terra and Aqua).	Aqua / Terra MODIS C6.1
sharper satellite (VIIRS)	VIIRS sees squares about 375 m wide, so it notices smaller fires. It has been flying since 2012.	S-NPP VIIRS 375 m
estimate on the sharper satellite's scale	What the sharper satellite would probably have counted. It is worked out from the older satellite, so it is an estimate, never a measurement.	estimated_viirs_reference_detections, harmonized
likely range (8 in 10)	We expect the true value to fall inside this range about 8 times out of 10.	80% interval
estimated past (before 2012)	Years before the sharper satellite existed. The numbers are estimates made from the older satellite.	backcast
years used to learn	Years the estimate was learned from, using months when both satellites were flying.	train split
years used to choose	Years held back to choose between different methods. Never used for learning.	validation split
years used to check	Years kept sealed until the end and used once to check the estimate honestly.	test split

Plain word	Meaning	Technical term
Checked on unseen years	The estimate beat both simple baselines on years it never saw, and its likely range was about right there.	validated_calibration
Raw comparison only	Real satellite counts are shown, but the estimate has not passed every check for this place, so treat it as a comparison.	observed_comparison
Demo data	Made-up numbers in the real file format, used only to test the app. They are not real observations.	ui_prototype
no data for this month	Nothing was recorded for this month. Missing is never shown as zero.	missing / unavailable
part of the month only	The satellite record covers only part of this month, so the count may be too low.	partial
unusually high / low for the season	Compared with the same month in other years, this month sits near the top or the bottom.	anomaly flag
data snapshot	A frozen copy of the data, so numbers do not change under you.	snapshot
simple baseline	A very simple rule used as a yardstick. A method only counts as good if it beats these.	B0_identity, B0_month_ratio
Copy the older satellite	One of the methods tried (chapter 5).	B0_identity
Statistical model	One of the methods tried (chapter 5).	B1_poisson
Tree model	One of the methods tried (chapter 5).	B2_xgb
Neural net	One of the methods tried (chapter 5).	D1_mlp

11. Credits

Parallax Fire by **Team Lumos**, created by **M Al Shahriar Hameem**.

Built for the **NASA Space Apps Challenge 2026**, challenge: *Harmonization of MODIS and VIIRS Hot Spots*.

The team page, with the team members, is at spaceappschallenge.org/2026/find-a-team/lumos2/.

Data courtesy of NASA FIRMS, NASA GIBS, CelesTrak, OpenFreeMap and OpenStreetMap contributors, Natural Earth and geoBoundaries (chapter 8).

Manual generated from snapshot `parallax-20261010T190934Z-ce88f489` on the numbers in its files; re-run `node scripts/manual/build_manual.mjs` to rebuild.